

Firms in Competitive Markets

Martin J. Conyon

Bentley University

The core text is N. Gregory Mankiw, *Principles of Microeconomics*.

- **Chapter 15** — Firms in Competitive Markets.

Read the chapter alongside these slides.

Where we are going

1. **What kind of market?** — four structures, and the one where the firm has no say over price.
2. **Revenue of a price taker** — why $P = AR = MR$.
3. **The output rule** — set marginal revenue against marginal cost, unit by unit.
4. **Three short-run scenarios** — profit, loss-but-operate, shut down.
5. **One rule, seen twice** — the shutdown rule *is* the supply curve.
6. **From firm to market** — adding firms up, and why long-run profit is zero.
7. **The dynamics of competition** — demand moves, profits appear, entry erases them.
8. **Beyond the baseline** — when the long-run supply curve tilts, and what competition is the benchmark for.

PART 1

What Kind of Market?

Four ways to organize a market

Structure	Sellers	Product	Firm's power over price
Perfect competition	many	identical	none
Monopolistic competition	many	differentiated	a little
Oligopoly	few	either	considerable
Monopoly	one	unique	the most

Everything here lives on the **top row**. The other three come later.

The fewer the sellers, the more each one can say about the price.

The three conditions

Competitive market. A market with (1) many buyers and many sellers, (2) identical goods across sellers, and (3) free entry and exit.

Our running firm: a maker of **standardized electronic connectors** — one of many firms stamping out an identical part.

- Many sellers: no firm is big enough to move the market.
- Identical product: buyers do not care whose connector they get.
- Free entry and exit: anyone may open a plant, or close one.

Price taker. A buyer or seller who accepts the market price as given and cannot influence it.

Suppose the market price of a connector is \$48.

- Charge \$49, and every customer walks to an identical rival: sales fall to **zero**.
- Charge \$47, and the firm gives away a dollar on every unit it could have sold at \$48 anyway.

The market sets the price. The firm chooses only its quantity.

The firm's three questions

With the price handed to it, the connector firm still decides:

1. **How much to produce?** — the output decision.
2. **Operate or shut down?** — when the price turns against it in the *short run*, while its fixed bill is unavoidable.
3. **Stay or exit?** — in the *long run*, when nothing is fixed and even the plant can be given up.

All three answers will come from **one comparison**: the market price against the firm's own cost curves.

Active Learning 1

Place each market on the spectrum — perfect competition, monopolistic competition, oligopoly, monopoly. For each one that is *not* perfectly competitive, name the condition it breaks. **The condition is the marks.**

- (a) corn farming
- (b) branded running shoes
- (c) home internet service in a town with two providers
- (d) the only pharmacy within fifty miles

Two minutes. Paper and pencil.

PART 2

Revenue of a Price Taker

Write the equation

The market price is $P = 48$. The firm can sell **any** quantity at that price. Its revenue is

$$TR = P \times q = 48q.$$

Average revenue. Revenue per unit sold: $AR = TR/q$.

Marginal revenue. The *change* in total revenue from selling one more unit: $MR = \Delta TR$.

Plug the numbers in

q	$TR = 48 \times q$	$AR = TR/q$	$MR = \Delta TR$
0	$48 \times 0 = 0$	—	—
1	$48 \times 1 = 48$	$48/1 = 48$	48
2	$48 \times 2 = 96$	$96/2 = 48$	48
3	$48 \times 3 = 144$	$144/3 = 48$	48
4	$48 \times 4 = 192$	$192/4 = 48$	48
5	$48 \times 5 = 240$	$240/5 = 48$	48
6	$48 \times 6 = 288$	$288/6 = 48$	48
7	$48 \times 7 = 336$	$336/7 = 48$	48
8	$48 \times 8 = 384$	$384/8 = 48$	48

The key property: $P = AR = MR$

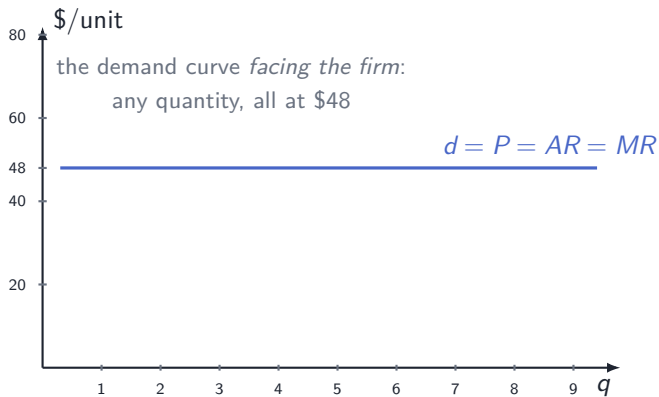
For a price taker, one number wears three hats:

- $AR = P$ always — for *any* firm, revenue per unit is the price.
- $MR = P$ **only because** selling one more unit never forces the price down.

For a price taker, the revenue from one more unit is the price itself.

A firm that must *cut* its price to sell more would find $MR < P$. Keep that thought for when we come to monopoly.

Draw the graph



Active Learning 2

A cherry grower sells at the market price of \$12 a crate — one grower among thousands.

- (a) Compute TR at 30 crates and at 31 crates. What is the MR of the 31st crate?
- (b) **Predict before computing:** she doubles her harvest to 60 crates. What happens to the market price? To her MR ?
- (c) A neighbor advises: “sell at \$11 to win customers.” What would she gain?
Explain why — the why is the marks.

Two minutes. Paper and pencil.

PART 3

The Output Rule

The cost side of the firm

The connector firm's total cost, in dollars per day:

$$TC = 144 + 36q - 8q^2 + q^3$$

- **Fixed cost** $FC = 144$ — owed even at $q = 0$.
- **Marginal cost** MC — the change in TC from one more unit: ΔTC .
- **Average variable cost** $AVC = VC/q$; **average total cost** $ATC = TC/q$.

These are the same four measures every firm has; here they come from *this* cost function.

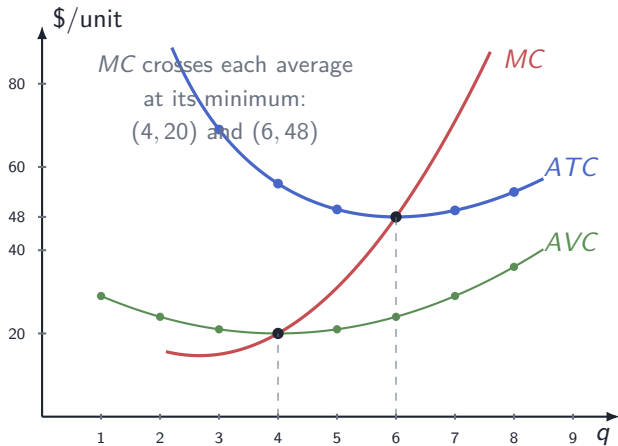
Plug the numbers in

$TC = 144 + 36q - 8q^2 + q^3$, row by row ($MC = \Delta TC$):

q	TC	VC	MC	AVC	ATC
0	144	0	—	—	—
1	173	29	29	29	173.0
2	192	48	19	24	96.0
3	207	63	15	21	69.0
4	224	80	17	20	56.0
5	249	105	25	21	49.8
6	288	144	39	24	48.0
7	347	203	59	29	49.57
8	432	288	85	36	54.0

Two minima to memorize: AVC bottoms at **\$20** ($q = 4$); ATC bottoms at **\$48** ($q = 6$).

Draw the graph



Set MR against MC , unit by unit

At $P = 48$, every unit brings in 48. What does each unit cost?

unit	MR	MC	$MR - MC$	verdict
1	48	29	+19	make it
2	48	19	+29	make it
3	48	15	+33	make it
4	48	17	+31	make it
5	48	25	+23	make it
6	48	39	+9	make it
7	48	59	-11	skip it
8	48	85	-37	skip it

The firm produces $q^* = 6$. (The six margins sum to \$144 — exactly the fixed cost.)

The rule

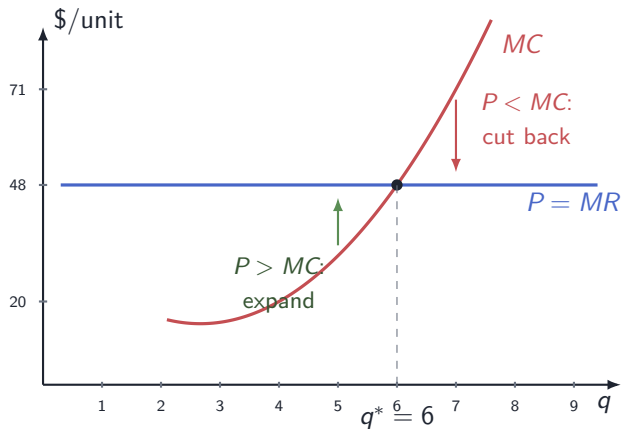
- While $MR > MC$: the next unit adds more than it costs — **expand**.
- While $MR < MC$: the last unit cost more than it added — **cut back**.

Produce where $MR = MC$.

This rule belongs to every profit-maximizing firm. For a price taker $MR = P$, so it wears a simpler outfit:

$$P = MC.$$

Draw the graph



Check it in totals

The marginal route said $q^* = 6$. Verify with total profit, $\pi = 48q - TC$:

q	$TR = 48q$	TC	π
4	192	224	-32
5	240	249	-9
6	288	288	0
7	336	347	-11
8	384	432	-48

Both routes agree: the peak is at $q = 6$. And the peak is **exactly zero** — because 48 is the minimum of ATC . *That number will run the whole second half of the lecture.*

Active Learning 3

The market price rises to $P = 50$.

- (a) **Predict before computing:** will the firm's best profit be positive or negative?
(Hint: $\min ATC = 48$.)
- (b) **Marginal route:** using the MC column (29, 19, 15, 17, 25, 39, 59, 85), find q^* .
- (c) **Total route:** compute $\pi = 50q - TC$ at $q = 5, 6, 7$ (TC : 249, 288, 347). **Do the two routes agree? Say why you checked.**

Four minutes. Paper and pencil.

PART 4

Three Short-Run Scenarios

The price is not always 48

Demand swings, and the market hands the firm a new price. Where the price lands against the **two trip-wires** decides everything:

Zone	Price	The firm...
1	$P > 48$ (min ATC)	operates at a profit
2	$20 \leq P < 48$	operates at a loss — on purpose
3	$P < 20$ (min AVC)	shuts down

One scenario per zone: $P = 71$, then $P = 31$, then $P = 15$.

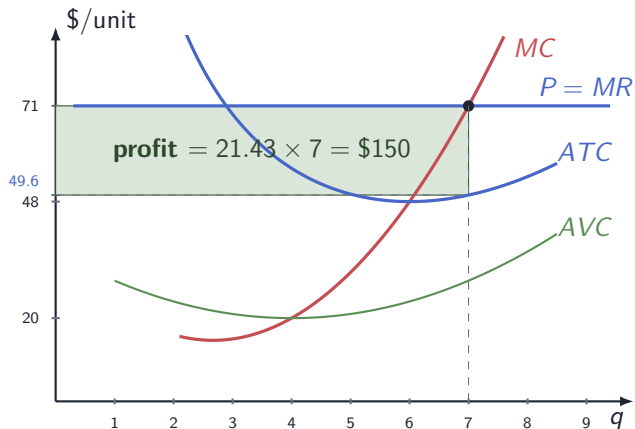
Scenario 1: $P = 71$ — profit

Marginal route: unit 7 costs $59 \leq 71$; unit 8 costs $85 > 71$. So $q^* = 7$.

Total route	$\pi = TR - TC = 71 \times 7 - 347 = 497 - 347 = +\150
Per-unit route	$\pi = (P - ATC) \times q = (71 - 49.57) \times 7 = 21.43 \times 7 = +\150

Two routes, same \$150 — and the per-unit route is exactly the **rectangle** we are about to draw.

Draw the graph: the profit rectangle



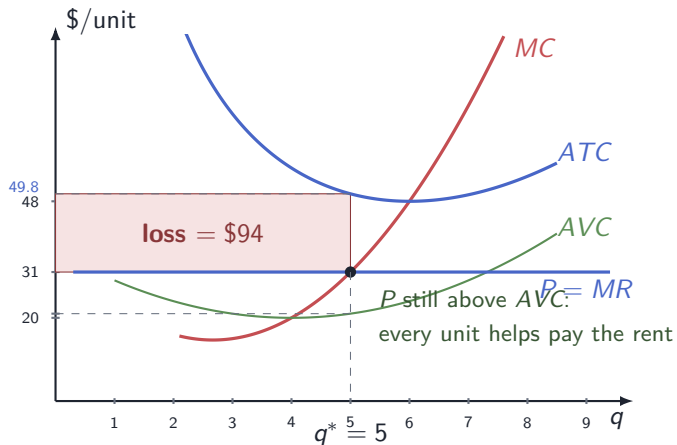
Scenario 2: $P = 31$ — a loss, taken on purpose

Marginal route: unit 5 costs $25 \leq 31$; unit 6 costs $39 > 31$. So $q^* = 5$, and $\pi = 31 \times 5 - 249 = -\94 . Why not quit?

	operate at $q = 5$	shut down
revenue	155	0
variable cost	105	0
fixed cost	144	144
result	-\$94	-\$144

−94 beats −144: revenue covers all the variable cost, with \$50 left toward the rent.

Draw the graph: the loss rectangle



Scenario 3: $P = 15$ — shut down

The cheapest the firm can *ever* make connectors, per unit of variable cost, is $AVC = 20$ (at $q = 4$).

At $P = 15$, **every** possible output sells below its variable cost: each unit produced deepens the loss. Best choice: $q = 0$, lose the fixed bill (\$144), and no more.

Shutdown rule. In the short run, operate if $P \geq \min AVC$; if $P < \min AVC$, produce zero. Fixed cost plays no part in the decision.

Why fixed cost stayed out of it: sunk costs

Sunk cost. A cost already committed that no current choice can recover.

In the short run the \$144 is owed **whatever the firm does** — operate or not. A cost that is the same across all options cannot help rank them.

The shutdown question is only: *does revenue beat variable cost?*

Money you cannot get back should not steer the decision ahead of you.

Active Learning 4

The price falls to exactly $P = 20$ — the bottom of AVC . At that price the firm's best positive output is $q = 4$ ($TC = 224$, $VC = 80$).

- (a) **Predict:** operate or shut down?
- (b) Compute the result of operating at $q = 4$.
- (c) Compute the result of shutting down.
- (d) What is special about this price? **The why is the marks — connect it to the number 20.**

Five minutes. Paper and pencil.

PART 5

One Rule, Seen Twice

Collect the answers

Every scenario asked the same question — *how much, at this price?* Line the answers up:

P	q supplied	from
15	0	below min AVC : shut
20	4 (or 0 — indifferent)	the knife edge
31	5	$P = MC$
48	6	$P = MC$
71	7	$P = MC$

A list of quantities offered at every price has a name: **a supply schedule**. We built the firm's supply curve without noticing.

Where the answers came from

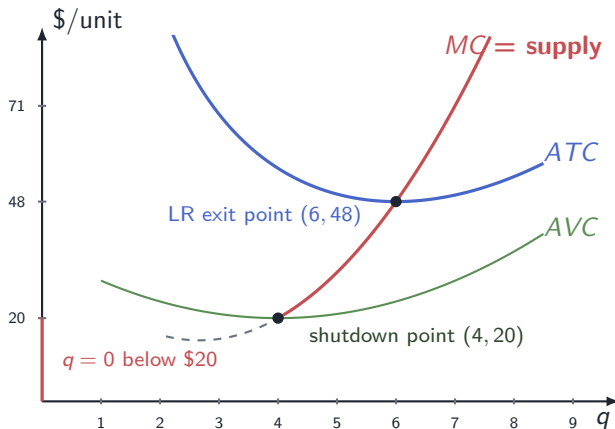
Every positive answer was read off the *MC curve*. Every zero came from the **shutdown rule**.

The firm's short-run supply curve *is* its *MC* curve above *AVC* — and zero below.

Not two facts. **One rule, seen twice:**

- asked at one price, it is the shutdown decision;
- asked at every price, it traces the supply curve.

Draw the graph



The long run: exit, not shutdown

Exit. Leaving the market for good: the plant is sold, the lease ends — *fixed* costs stop too.

In the long run **nothing is sunk**. The comparison is no longer P vs AVC but P vs ATC — all costs must be covered.

- Exit if $P < \min ATC = 48$: the business cannot pay its whole way.
- An outsider **enters** if $P > 48$: the business more than pays its way.

The firm's *long-run* supply curve: MC above ATC .

The decision rules, one table

Price	Short run	Long run
$P < 20$	shut down	exit
$20 \leq P < 48$	operate at a loss	exit
$P = 48$	operate, $\pi = 0$	stay — and no one enters
$P > 48$	operate, $\pi > 0$	stay — and entrants come

Short run: P against **min** AVC (can revenue beat variable cost?).

Long run: P against **min** ATC (can revenue beat *all* cost?).

Active Learning 5

For each price, give the firm's **short-run** action and its **long-run** action (min $AVC = 20$, min $ATC = 48$):

$$P = 15, \quad P = 35, \quad P = 45, \quad P = 55.$$

Then, in **one sentence**: why are the shutdown rule and the supply curve the *same* fact and not two?

Three minutes. Paper and pencil.

PART 6

From Firm to Market

Add the firms up

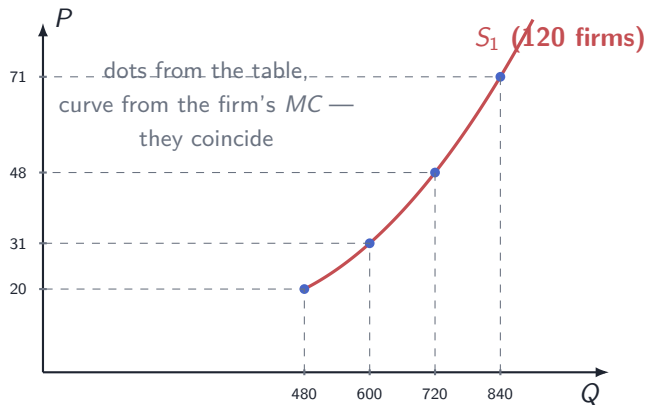
Short run: the cast is fixed — **120 identical firms**, no one can enter or leave yet.

Market supply at each price is $120 \times q$:

P	each firm's q	market $Q^s = 120 \times q$
20	4	$120 \times 4 = 480$
31	5	$120 \times 5 = 600$
48	6	$120 \times 6 = 720$
71	7	$120 \times 7 = 840$

The market supply curve is the firm's, stretched 120 times wider.

Draw the graph



The long run: profit moves the cast

In the long run the number of firms is **not** fixed — and profit is what changes it.

- $P > 48$: incumbents profit $\Rightarrow$ **entry** $\Rightarrow$ supply shifts right $\Rightarrow P$ falls.
- $P < 48$: incumbents lose $\Rightarrow$ **exit** $\Rightarrow$ supply shifts left $\Rightarrow P$ rises.

Long-run competitive equilibrium. The state where the traffic stops: $P = MC$ (each firm maximizes), $P = \min ATC$ (zero economic profit), and each firm produces at its efficient scale — three statements, one point.

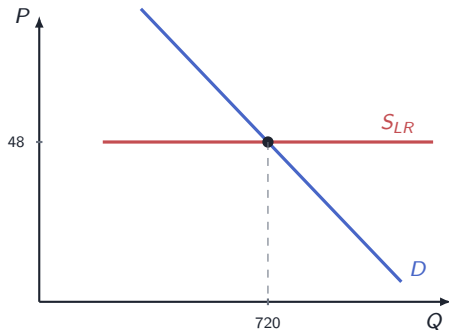
Why stay in business for zero profit?

The puzzle from Part 3, resolved. **Economic** cost counts *everything* given up — including the owner's next-best salary and the return their capital could earn elsewhere.

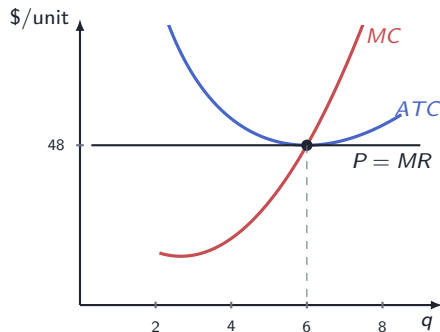
Zero economic profit	revenue covers all costs <i>including</i> those implicit ones
The accountant's books	show a healthy positive profit — implicit costs excluded

Zero economic profit = a normal return: the owners do exactly as well as their best alternative.

Firm and industry, in equilibrium together



Market: S_{LR} flat at 48 — the firm count adjusts



Each firm: $q = 6$ at min ATC — zero profit

$720 = 120 \times 6$: the market total is the head count times the efficient scale.

Active Learning 6

A founder runs a connector firm. Her accountant reports an annual profit of **\$80,000**. She gave up a \$70,000 job to run it, and she has \$300,000 of savings tied up in the plant, which could earn 5% elsewhere.

- (a) **Predict:** the books look healthy — is she doing well?
- (b) Compute her *economic* profit.
- (c) Long run: stay or exit? **Reasoning is the marks.**

Three minutes. Paper and pencil.

PART 7

The Dynamics of Competition

What does a market do when demand moves?

The static picture is done. Now the real power of competition: it **repairs itself**. Three stages, every one computed:

1. **Rest.** Long-run equilibrium: zero profit, 120 firms.
2. **Shock.** Demand jumps; the price spikes; profits appear *overnight*.
3. **Repair.** Profit summons entrants; entry pushes the price back down — and we will compute *exactly* where it stops.

Then the same film **in reverse**: demand falls, losses, exit.

Stage 1: the resting market, in numbers

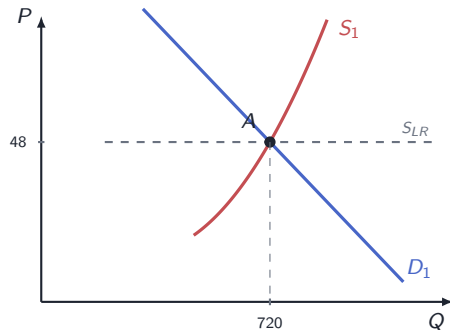
Demand: $Q^d = 1200 - 10P$. Test the zero-profit price $P = 48$:

demand	$1200 - 10 \times 48 = 1200 - 480 = 720$
supplied	$120 \text{ firms} \times 6 = 720$

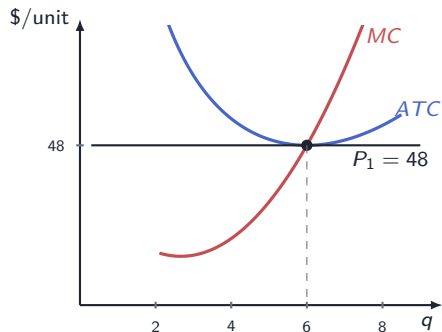
Balance. Call this point **A**: $P = 48$, $Q = 720$, 120 firms, each at $q = 6$, $\pi = 0$.

No one profits, no one loses; **no one enters, no one leaves**.

Stage 1: draw the pair



Market: at rest at $A = (720, 48)$



Firm: $q = 6$, price kisses min ATC — zero profit

Stage 2: demand jumps

A new technology needs connectors everywhere. Demand shifts to $Q^d = 1550 - 10P$.

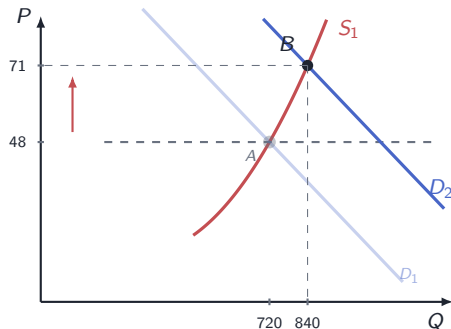
At the old price $P = 48$: demanded = $1550 - 480 = 1070$, supplied = 720. **Shortage of 350 — the price must rise.**

Find the new short-run balance (cast still frozen at 120):

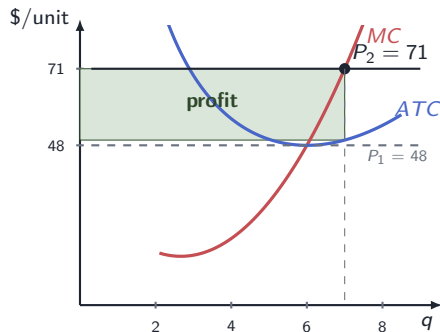
try P	demanded	supplied (120 firms)
48	1070	$120 \times 6 = 720$ — shortage
71	$1550 - 710 = \mathbf{840}$	$120 \times 7 = \mathbf{840}$ — balance

Point **B**: $P = 71$, $Q = 840$.

Stage 2: draw the pair



Market: D jumps; price spikes to 71 at B



Firm: up MC to $q = 7$ — profit appears

Stage 2: the profit, in numbers

Each of the 120 firms, at $P = 71$, $q = 7$:

Total route	$\pi = 71 \times 7 - 347 = 497 - 347 = +\150
-------------	--

Per-unit route	$\pi = (71 - 49.57) \times 7 = +\150
----------------	--

\$150 per firm, every day — earned for *being there when demand arrived*.

To an outsider, \$150 a day is not a number. It is an invitation.

Stage 3: profit is an invitation

Entrants build plants and start producing. Each arrival shifts market supply **right** — and pushes the price **down**.

Snapshot partway through entry, price down to $P = 60$: each firm still picks $q = 7$ (unit 7 costs $59 \leq 60$), and

$$\pi = 60 \times 7 - 347 = 420 - 347 = +\$73.$$

Still positive — **so entry has not finished**. More come; the price keeps sliding.

Entry continues as long as the price leaves a profit. So where, exactly, does it stop?

Why entry stops — exactly

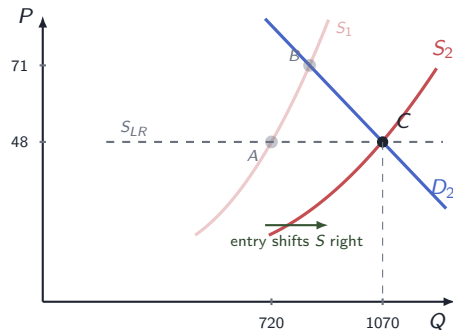
At $P = 48$, demand is $1550 - 480 = 1070$; each firm at efficient scale makes 6. But $1070/6 = \mathbf{178.33}$ — and a third of a firm cannot enter. **So check the whole numbers on either side:**

	$Q = 6n$	$P = (1550 - Q)/10$	π per firm $= 6P - 288$
$n = 178$	1068	\$48.20	$289.20 - 288 = +\$1.20$
$n = 179$	1074	\$47.60	$285.60 - 288 = -\$2.40$

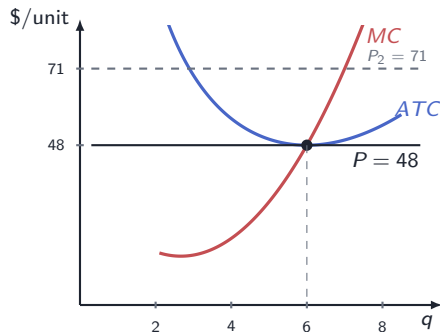
The 178th firm still covers all its costs. A 179th would push the price to \$47.60 and lose — **so it stays out.**

Entry stops at 178 firms: at the last entrant that still breaks even.

Stage 3: draw the pair



Market: entry drives price back to 48 at C



Firm: back to $q = 6$ — the rectangle is gone

A to C: what changed, and what did not

	A (before)	C (after)	
price	48	48	unchanged
each firm's q	6	6	unchanged
each firm's π	0	0	unchanged
market Q	720	1070	+350
number of firms	120	178	+58

The demand boom was absorbed entirely by *new firms* — not by a permanently higher price.

Now run the film backward: demand falls

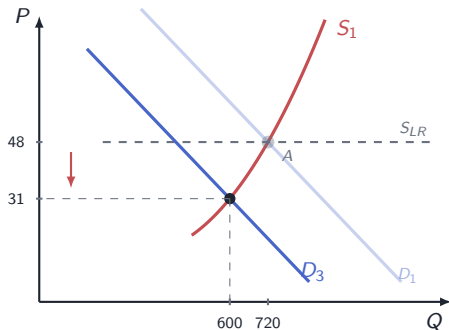
From rest at A (120 firms), demand *falls* to $Q^d = 910 - 10P$.

At $P = 48$: demanded = $910 - 480 = 430$, supplied = 720. **Glut of 290 — the price must fall.**

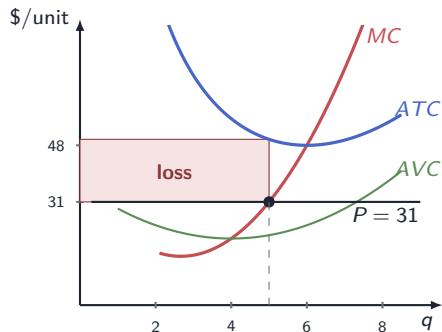
try P	demanded	supplied (120 firms)
31	$910 - 310 = \mathbf{600}$	$120 \times 5 = \mathbf{600} — \mathbf{balance}$

Each firm: $\pi = 31 \times 5 - 249 = -\94 — scenario 2 come to life. Since $31 > 20 = \min AVC$: **operate at a loss** — and plan to leave.

Draw the pair: losses



Market: D falls; price sinks to 31



Firm: $q = 5$, P above AVC — bleed, don't quit

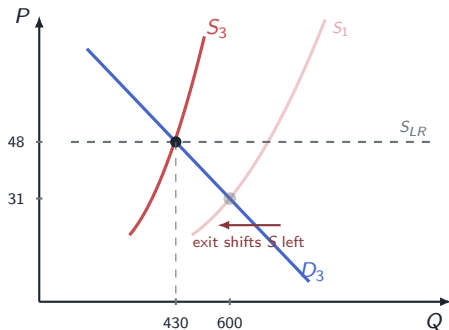
Exit — and where it stops

Losses push firms out; each exit shifts supply **left**; the price **recovers**. At $P = 48$, demand is $910 - 480 = 430$, and $430/6 = 71.67$. **Whole firms again — check both sides:**

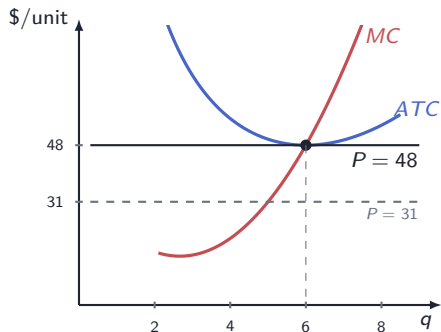
$Q = 6n$		$P = \frac{910 - Q}{10}$	$\pi \text{ per firm} = 6P - 288$
$n = 72$	432	\$47.80	$286.80 - 288 = -\$1.20$ — still losing: <i>one more leaves</i>
$n = 71$	426	\$48.40	$290.40 - 288 = +\$2.40$ — covered: <i>exit stops</i>

Same machine, opposite gear: losses expel firms until the survivors break even.

Draw the pair: recovery



Market: exit lifts price back to 48; $Q = 430$



Surviving firm: back to $q = 6$, zero profit

Three results to keep

1. **Short run:** a demand shift moves the *price*, and with it profits (+\$150) or losses (−\$94).
2. **Long run:** entry and exit grind economic profit back to *zero* — stopping at the last whole firm that breaks even.
3. **The adjustment lives in the number of firms** ($120 \rightarrow 178$, or $120 \rightarrow 71$), *not* in the price, which returns to $\min ATC = 48$.

Competition converts temporary profit into permanent capacity — and temporary loss into permanent downsizing.

Active Learning 7 — the whole machine

The market rests at A : $P = 48$, 120 firms, $Q = 720$. Demand shifts to

$$Q^d = 1310 - 10P.$$

- (a) **Predict, no computing:** short run — which way do price and profit move? Long run — what is the price?
- (b) Long-run quantity at that price?
- (c) **Number of firms** — do the whole-firm check on both sides, as we did for the boom.
- (d) Who produces the extra output — old firms or new?

Five minutes. Paper and pencil.

PART 8

Beyond the Baseline

When the flat line tilts (1): scarce inputs

The horizontal S_{LR} leaned on a quiet assumption: **input prices hold still as the industry grows.**

Suppose connector plants need a specialized alloy in limited supply. Entry during a boom **bids up the alloy's price** — so every firm's cost curves shift up, and min ATC rises with the size of the industry.

Then a bigger market needs a *permanently* higher price to cover the higher cost:

the long-run supply curve slopes upward — though still flatter than short-run supply.

When the flat line tilts (2): unequal firms

The other quiet assumption: **all firms, current and potential, share the same costs.**

Suppose instead the best sites and managers are taken: entrants' min ATC exceeds incumbents'. Then the price must rise to the **marginal entrant's** break-even before entry stops — and low-cost incumbents keep a *persistent* surplus.

That surplus is not entry-proof profit but **rent** to the scarce advantage — the prime site, the rare talent — and it survives in long-run equilibrium.

What competition delivers

At the long-run rest point, three things are true at once — none of them decreed by anyone:

- $P = MC$: every unit worth at least its production cost to a buyer **gets made** — none worth less does.
- $P = \min ATC$: total output is produced at the **cheapest possible cost per unit**.
- $\pi = 0$: consumers pay **no more than the product truly costs** — including a normal return to the owners.

This is the benchmark. Every other market structure is measured by its distance from it.

The other pole: one seller

A **monopoly** is a market served by a single firm, protected from entry. It faces the *whole* downward-sloping market demand — to sell more it must **cut the price**, so its $MR < P$.

	Competition	Monopoly
sellers	many	one
demand facing firm	horizontal at P	the market's, sloping down
revenue logic	$MR = P$	$MR < P$
output rule	$P = MC$	$MR = MC$, so $P > MC$
long-run π	zero (entry)	can persist (entry blocked)

When we come to monopoly, the right-hand column is the whole story.

Active Learning 8

In a farming market, would-be entrants all face $\min ATC = \$48$ (at $q = 6$). A few incumbent farms sit on unusually good land: their $\min ATC = \$40$, also at $q = 6$.

- (a) What is the long-run price? **Whose costs pin it down — and why?**
- (b) Compute a good-land farm's long-run profit per day.
- (c) Why doesn't entry compete that profit away?

Three minutes. Paper and pencil.

Competition is a process, not a state.

Profit is the signal, entry is the response,
and the process rests only where the marginal firm
just breaks even — at $P = MC = \min ATC$.